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The Economic Effects of Public Financing: Evidence from Municipal Bond Ratings Recalibration

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1 Big picture

- **Question.** Do financial constraints of local governments affect local employment, income, and growth?
- **Shock.** The paper exploits Moody's April–May 2010 recalibration of U.S. municipal bond ratings. The recalibration upgraded many municipal ratings by zero to three notches without reflecting any change in issuer credit quality.
- **Identification idea.** Upgraded local governments receive an exogenous relaxation of public-sector financing constraints. Non-upgraded governments and counties form the control group.
- **Setting.** The shock affected nearly 70,000 municipal bond issues, about 18,000 local governments, and roughly \$2.2 trillion of par value.
- **First-stage bond-market result.** Upgraded local governments receive higher Moody's ratings on new issues after recalibration, issue more debt, and face lower offer yields. S&P ratings do not move, which supports the interpretation that Moody's changes are scale changes rather than credit-quality changes.
- **Public spending result.** Upgraded local governments raise expenditures after their ratings improve. A one-standard-deviation increase in the county fraction of upgraded local governments raises local government expenditures by about 0.5%.
- **Public employment result.** Local government employment also rises by about 0.5% for a one-standard-deviation increase in county exposure to recalibration.
- **Private-sector spillover.** Private employment rises by about 0.3%. The effect is concentrated in nontradable industries and education, not in tradable industries.
- **Income result.** County-level income rises by about 0.5% for a one-standard-deviation increase in exposure.
- **Multiplier estimates.** Using the recalibration as an instrument for local government expenditures, the paper estimates that \$1 million of local government spending creates about 51 jobs, 45 outside the public sector. The implied cost per job is about \$20,000 per year, and the local income multiplier is about 1.9.
- **Heterogeneity.** Effects are stronger for counties with high unemployment and for financially constrained local governments, especially high-leverage local government units.
- **Economic takeaway.** Municipal bond markets do not only determine financing costs. During recessions, easing local governments' access to debt can raise public spending and generate measurable private-sector employment and income gains.

2 Data and measurement

2.1 Moody's 2010 municipal ratings recalibration

- Before 2010, Moody's used a municipal rating scale that measured distance to distress, while its global scale measured expected losses.
- The municipal scale was more conservative than the global scale and often delivered lower ratings than S&P for the same municipal issue.
- In March 2010, Moody's announced that it would recalibrate municipal ratings to the global scale.
- The recalibration took place over April–May 2010 and generated zero-to-three-notch upgrades.
- Moody's explicitly stated that the recalibration did not reflect improved issuer credit quality or a changed credit opinion.

Interpretation: The recalibration is useful because it changes ratings mechanically. It relaxes financing constraints without being caused by local economic fundamentals.

2.2 Bond market data

- The paper obtains the recalibrated bond list from Moody's.
- Municipal primary-market data come from the Ipreo i-Deal new issues database.
- The bond-market sample covers April 2006 to March 2013: four years before and three years after the recalibration.
- The sample is restricted to local governments that issued bonds during the four years before the recalibration.
- The analysis excludes state-level bonds because state bonds cannot be assigned to a particular county.
- Insured bonds are excluded because insured-bond ratings reflect the insurer's credit quality rather than the municipality's credit quality.
- Bond outcomes include Moody's rating, S&P rating, issue amount, and offer yield.

Interpretation: The bond-market sample is designed to measure how the recalibration affected local governments' access to municipal debt markets.

2.3 County-level treatment variable

Let

$$Recalibrated_i$$

be the fraction of local government units in county i whose outstanding bonds were upgraded by Moody's recalibration.

- A local government is treated if at least one of its outstanding bonds is upgraded in the recalibration.
- County-level exposure is continuous because a county may contain many local government units, only some of which are upgraded.
- The main county sample contains counties in which at least one local government issued bonds before the recalibration.

Interpretation: The continuous treatment captures the intensity of exposure to the credit-supply shock at the county level.

2.4 Economic outcome data

- Local government expenditures are from the Census Bureau’s Annual Survey of State and Local Government Finances.
- Local government employment is from the Census Bureau’s Government Employment and Payroll Survey.
- Private-sector employment by industry and county is from County Business Patterns.
- Quarterly employment robustness uses Quarterly Workforce Indicators.
- County income is from IRS Statistics of Income and is measured as total wages and salaries.
- House price controls come from FHFA house price indexes.
- County household counts come from Census summary files.

Interpretation: The real outcomes measure both direct local-government responses and spillovers into the private sector.

2.5 Sector definitions

- Tradable employment is manufacturing employment, NAICS 31–33.
- Nontradable employment is retail, food, and accommodation employment, NAICS 44–45 and 72.
- Construction and education are studied separately because public spending may directly affect infrastructure and schools.

Interpretation: Sector heterogeneity is central to the multiplier interpretation. A local demand shock should show up more strongly in nontradable sectors than in tradable sectors.

2.6 Main samples

- The baseline county sample covers counties with at least one pre-recalibration local bond issuer.
- The urban-county sample restricts to counties with more than 50,000 people.
- The urban sample improves comparability because treatment counties tend to be larger before controls.
- Main regressions include state-by-year fixed effects and often county-size-decile-by-year fixed effects to address regional and size-specific shocks.

Interpretation: Treatment counties are larger on average, but adjusted differences using state-year and size-year fixed effects are much smaller. The empirical design builds these adjustments into the regressions.

3 Models and empirical specifications

3.1 Bond-market first stage: ratings

The paper estimates a difference-in-differences specification:

$$Rating_{it} = \beta_1 Recalibrated_i \times Post_t + \beta_2 X_{it} + \eta_i + \eta_{state,t} + \eta_{size,t} + \varepsilon_{it}. \quad (1)$$

- $Rating_{it}$ is the average rating for new issues in county i and event year t .
- $Post_t = 1$ after the April–May 2010 recalibration.
- η_i are county fixed effects.
- $\eta_{state,t}$ are state-by-year fixed effects.
- $\eta_{size,t}$ are county-size-decile-by-year fixed effects.
- X_{it} includes issue-level controls such as GO/revenue status, Build America Bond status, and duration.

Interpretation: A positive β_1 for Moody’s but not for S&P means the recalibration changed Moody’s scale rather than fundamentals.

3.2 Bond-market first stage: issuance and yields

The same design is applied to:

$$\log(IssueAmount_{it}) \quad \text{and} \quad OfferYield_{it}.$$

Interpretation: More issuance and lower yields indicate that ratings upgrades relax local governments’ financing constraints and expand debt capacity.

3.3 Real-outcome reduced form

For real outcomes Y_{it} such as government expenditures, government employment, private employment, sector employment, and income:

$$Y_{it} = \beta Recalibrated_i \times Post_t + \Gamma' Controls_{it} + \eta_i + \eta_{state,t} + \eta_{size,t} + \varepsilon_{it}. \quad (2)$$

- Outcomes are generally logarithms.
- Controls include house price changes and number of households.
- Standard errors are clustered at the county level.

Interpretation: The coefficient β is the county-level reduced-form effect of exposure to the recalibration on local economic outcomes.

3.4 Local-government-level specification

At the local government level, the paper uses:

$$Y_{gct} = \beta RecalibratedDummy_g \times Post_t + \Gamma' Controls_{gct} + \eta_g + \eta_{county,t} + \eta_{type,t} + \varepsilon_{gct}. \quad (3)$$

- g indexes local government units.
- $RecalibratedDummy_g = 1$ if the local government received an upgrade in the recalibration.
- Fixed effects can include local government, county-year, and local-government-type-year fixed effects.

Interpretation: The local-government specification asks whether treated local governments themselves increase expenditures or employment, while the county-level specification captures spillovers within the county.

3.5 Fiscal multiplier IV

To estimate multipliers, the paper instruments local government expenditures with the recalibration exposure:

$$\log(GovExp_{it}) = \pi Recalibrated_i \times Post_t + FE + Controls + u_{it}, \quad (4)$$

$$Y_{it} = \theta \log(\widehat{GovExp}_{it}) + FE + Controls + \varepsilon_{it}. \quad (5)$$

- Y_{it} is government employment, private employment, or income.
- The urban-county sample has first-stage F -statistics above ten in Table 10.
- Elasticities are converted into dollar or job multipliers using average ratios of employment and income to government expenditures.

Interpretation: This turns the reduced-form recalibration shock into an estimate of how much employment or income changes per dollar of local government spending.

3.6 Heterogeneity specifications

The paper estimates triple-interaction models:

$$Y_{it} = \beta_1 Recalibrated_i \times Post_t + \beta_2 Recalibrated_i \times Post_t \times HighSlack_i + FE + Controls + \varepsilon_{it}, \quad (6)$$

and similarly for high leverage:

$$Y_{gct} = \beta_1 RecalibratedDummy_g \times Post_t + \beta_2 RecalibratedDummy_g \times Post_t \times HighLeverage_g + FE + Controls + \varepsilon_{gct}. \quad (7)$$

Interpretation: Higher effects in high-slack counties support a Keynesian demand-channel interpretation. Higher effects for high-leverage governments support the financial-constraint interpretation.

4 Identification and empirical design

4.1 What identifies the main effect

- Moody’s recalibration generates exogenous variation in municipal ratings.
- The recalibration was a scale alignment, not a credit-quality upgrade.
- The difference-in-differences design compares upgraded and non-upgraded local governments before and after April–May 2010.
- County fixed effects absorb time-invariant local differences.
- State-by-year fixed effects absorb statewide shocks, transfers, pensions, and policy changes.
- County-size-decile-by-year fixed effects absorb the possibility that large and small counties recovered differently after the Great Recession.

Interpretation: The identifying assumption is that, conditional on these controls, treated and control counties would have followed similar post-2010 trends absent the recalibration.

4.2 Why the S&P placebo matters

- If the recalibration reflected true credit-quality improvements, S&P ratings should also rise for the same issuers.
- The paper finds that Moody’s ratings increase after recalibration, while S&P ratings do not.

Interpretation: This placebo supports the claim that the shock is a Moody’s scale change rather than an economic fundamental shock.

4.3 Addressing treatment-control imbalance

- Treated counties are larger on average before adjustment.
- Table 2 shows that state-year and size-year adjustments remove much of this difference.
- The paper also uses an urban county sample, which makes counties more comparable.

Interpretation: The design explicitly tackles the concern that large counties were treated and recovered differently.

4.4 Addressing Great Recession confounds

- The event occurs during the post-crisis period, when counties were recovering unevenly.
- Controls include house prices and households.
- State-by-year and size-year fixed effects absorb broad recovery patterns.
- Robustness checks include group-specific trends, urban counties, counties with multiple issuers, counties with larger bond issuance, and all counties with outcome data.

Interpretation: The paper tries to isolate a credit-supply shock to local governments from broader recession and recovery dynamics.

4.5 Why sector results matter

- Nontradable sectors depend on local demand.
- Tradable sectors depend more on national or global demand.
- The paper finds effects concentrated in nontradable employment and education, not tradable employment.

Interpretation: The sector pattern is consistent with local government spending creating local demand rather than merely reflecting broad economic recovery.

4.6 What the paper cannot fully prove

- Moody's recalibration is not randomized at the county level.
- The treatment variable reflects the fraction of local governments upgraded, so exposure intensity is partly shaped by local government structure.
- Some counties may be exposed to other policies that coincide with the Great Recession recovery.
- The estimates are local multipliers, not national multipliers, because cross-county spillovers and general equilibrium effects are not fully captured.

Interpretation: The evidence is persuasive because the shock is plausibly exogenous and multiple outcomes move coherently, but the estimates should be read as local reduced-form effects and local IV multipliers.

5 Tables and figures

5.1 Table 1 and Figure 1: sample, treatment exposure, and geography

Read:

- Table 1 shows that baseline counties issue an average of \$170 million in bonds per year, but the distribution is highly skewed, with a median of \$15 million.
- Local government expenditures average \$766 million in the bond-issuer sample, while private employment averages about 60,170 workers.
- The main treatment variable, *Recalibrated*, has a mean of 0.048 and a standard deviation of 0.094 in the bond-issuer sample.
- The urban-county sample is larger on average and has a higher mean treatment intensity of 0.066.
- Figure 1 shows that recalibrated counties are geographically spread across the United States rather than concentrated in one region.

Economic message: Treatment exposure is widespread and not simply a single-region event. The summary statistics also show why size controls are important.

Table 1
Summary statistics

	Mean	Median	SD	Minimum	Maximum	No. of observations	No. of counties
<i>A. County-level variables, sample of bond issuers</i>							
Moody's ratings	17.86	18.00	1.70	12.00	21.00	4,872	1,319
S&P ratings	18.61	19.00	1.80	12.00	21.00	2,147	608
Issue amount (\$ million)	170.1	15.3	1,482.7	0.1	66,400.0	6,843	1,859
Offer yield (%)	2.83	3.03	1.67	0.10	6.60	6,843	1,859
Households (thousand)	53.28	16.41	134.66	0.29	3,133.77	12,348	1,768
Local government expenditures (\$ million)	765.7	143.1	3,445.8	0.1	108,487.3	12,180	1,740
Local government employment (thousand)	6.20	1.37	20.44	0.00	437.54	8,067	1,154
Private employment (thousand)	60.17	13.59	173.09	0.01	3,910.43	12,346	1,768
Tradable employment (thousand)	4.39	0.75	15.80	0.00	417.55	8,405	1,407
Nontradable employment (thousand)	14.02	3.57	35.87	0.01	778.39	12,314	1,766
Construction employment (thousand)	3.09	0.67	8.56	0.00	171.09	12,368	1,768
Education employment (thousand)	1.75	0.06	7.20	0.00	140.50	11,313	1,676
Income (\$ million)	2,924.3	652.6	8,106.7	5.7	197,206.3	12,348	1,768
Recalibrated	0.048	0.000	0.094	0.000	1.000	12,374	1,768
<i>B. County-level variables, sample of urban counties</i>							
Moody's ratings	18.25	18.39	1.61	12.00	21.00	3,168	701
S&P ratings	18.81	19.00	1.61	12.00	21.00	1,918	485
Issue amount (\$ million)	218.5	45.1	938.8	0.1	27,388.6	3,912	799
Offer yield (%)	2.80	3.05	1.70	0.10	6.60	3,912	799
Households (thousand)	93.62	43.39	176.76	1.52	3,133.77	6,501	932
Local government expenditures (\$ million)	1,400.2	462.3	4,678.8	1.0	108,487.3	6,363	909
Local government employment (thousand)	9.28	3.81	24.63	0.00	437.54	5,321	761
Private employment (thousand)	107.97	42.27	228.00	0.16	3,910.43	6,522	932
Tradable employment (thousand)	6.23	1.41	18.84	0.00	417.55	5,741	879
Nontradable employment (thousand)	24.85	11.63	46.72	0.03	778.39	6,519	932
Construction employment (thousand)	5.68	2.28	11.26	0.01	171.09	6,439	932
Education employment (thousand)	4.06	0.93	10.61	0.00	140.50	4,871	783
Income (\$ million)	5,301.9	2,088.0	10,642.4	35.4	197,206.3	6,501	932
Recalibrated	0.066	0.020	0.115	0.000	1.000	6,532	933

This table shows mean, median, standard deviation, minimum, maximum, and number of observations for each variable. The sample in panel A consists of counties in which at least one local government issued bonds in the municipal bond market during the four-year period before the recalibration (April 2006–March 2010). The

Table 1: Summary statistics

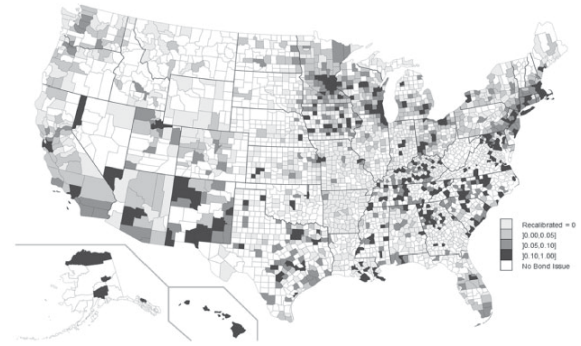


Figure 1
Local government units upgraded due to the recalibration by county
The map shows the fraction of local government units in a given county upgraded during the Moody's recalibration (*Recalibrated*). Counties in white have no local government unit that issued bonds during the four-year period before the recalibration (April 2006–March 2010) (1,373 counties). Counties in light grey have no upgraded local government unit (811 counties). Counties in medium grey, dark grey, and black are in the bottom tercile (327 counties), medium tercile (313 counties), and top tercile (317 counties) of the distribution of the *Recalibrated* variable (considering nonzero values), respectively.

Figure 1: Upgraded local government units by county

5.2 Table 2: pretreatment balance

Read:

- Treated counties are larger in raw pretreatment levels: for example, households average 82,120 in treated counties versus 19,410 in control counties.
- After adjustment for state-year and county-size-decile-year effects, the household and private employment differences are no longer economically large.
- Pretreatment growth rates in private employment and income are very similar across treatment and control groups.
- The adjusted difference in offer yield remains positive, suggesting treated issuers initially faced somewhat higher borrowing costs.

Economic message: Table 2 motivates the paper’s empirical specification. Treated and control counties are not identical in raw levels, so fixed effects and size-year controls are essential.

Table 2
Treatment and control groups pretreatment characteristics

A. County-level variables, sample of bond issuers

	Mean			Difference (adjusted)	Number of observations	
	Recalibrated >0	Recalibrated = 0	Difference		Recalibrated >0	Recalibrated = 0
Moody’s ratings	17.41	17.58	−0.17*	−0.05	1,291	1,498
S&P ratings	18.66	18.29	0.37**	0.32*	651	576
Issue amount (\$ million)	254.6	104.8	149.8	41.0	1,459	2,483
Offer yield (%)	3.19	2.97	0.22***	0.29**	1,459	2,483
Households (thousands)	82.12	19.41	62.71***	6.86	2,895	2,868
Private employment (thousands)	97.52	18.62	78.90***	8.42	2,898	2,866
Fraction of local government employment	0.059	0.052	0.006***	0.005*	2,298	2,278
Growth local government expenditures	0.054	0.050	0.004	0.007*	1,904	1,886
Growth local government employment	0.011	0.019	−0.008*	0.000	1,542	1,518
Growth local private employment	−0.026	−0.029	0.003*	0.000	1,933	1,910
Growth income	−0.008	−0.009	0.001	0.000	1,930	1,912
Local government expenditures (\$ million)	1,169.1	287.5	881.6***	−37.6	1,904	2,829
Local government employment (thousand)	8.19	2.67	5.52***	−0.47	1,542	2,278
Private employment (thousand)	97.52	18.62	78.90***	8.42	1,933	2,866
Tradable employment (thousand)	6.60	1.10	5.49***	1.43**	1,904	2,388
Nontradable employment (thousand)	22.43	4.32	18.11***	2.80**	1,542	2,862
Construction employment (thousand)	5.69	0.98	4.71***	0.78**	1,933	2,868
Education employment (thousand)	2.48	0.54	1.94***	−0.08	1,933	2,779
Income (\$ million)	4,442.7	831.7	3,611.0***	456.5	1,930	2,868

B. County-level variables, sample of urban counties

Moody’s ratings	17.87	17.99	−0.12	−0.12	873	927
S&P ratings	18.83	18.60	0.23	0.13	498	575
Issue amount (\$ million)	213.4	198.8	14.7	−5.3	1,203	1,025
Offer yield (%)	3.10	2.79	0.31***	0.31**	1,203	1,025
Households (thousands)	124.3	40.4	83.9***	12.8	1,776	1,020
Private employment (thousands)	150.0	40.4	109.6***	21.0	1,776	1,020
Fraction of local government employment	0.064	0.063	0.001	0.004	1,654	624
Growth local government expenditures	0.051	0.045	0.006	0.011**	1,176	642
Growth local government employment	0.012	0.011	0.001	0.010*	1,104	416
Growth local private employment	−0.026	−0.023	−0.004	−0.004	1,184	680
Growth income	−0.008	−0.005	−0.002	0.002	1,184	680
Local government expenditures (\$ million)	1,801.4	666.2	1,135.2***	−117.2	1,764	963
Local government employment (thousand)	11.00	5.35	5.65**	−0.58	1,657	624
Private employment (thousand)	150.00	40.35	109.65***	21.03	1,776	1,020
Tradable employment (thousand)	9.00	1.67	7.33***	1.96**	1,707	777
Nontradable employment (thousand)	34.13	9.22	24.91***	6.74***	1,776	1,017
Construction employment (thousand)	8.80	2.25	6.55***	2.10***	1,776	1,020
Education employment (thousand)	3.83	1.24	2.59***	−0.10	1,771	980
Income (\$ million)	6,863.5	1,862.7	5,000.8***	1,070.2**	1,776	1,020

This table shows pretreatment means and *p*-values of differences in means (raw and adjusted) between treatment and control groups. The differences are adjusted by state-year fixed effects and by county-size decile-by-year fixed effects. The sample in panel A consists of counties in which at least one local government issued bonds in the municipal bond market during the four-year period before the recalibration (April 2006–March 2010). The sample

5.3 Table 3 and Figures 2–3: Moody’s ratings move, S&P ratings do not

Read:

- Table 3 shows that county-level Moody’s ratings rise by about 0.70–0.73 notches for treated counties after the recalibration.
- At the local-government level, the Moody’s effect is about 0.65–0.67 notches.
- S&P rating changes are small and statistically insignificant.
- Figure 2 shows a clear upward jump in Moody’s ratings for treated issuers after the event.
- Figure 3 shows no comparable S&P response.

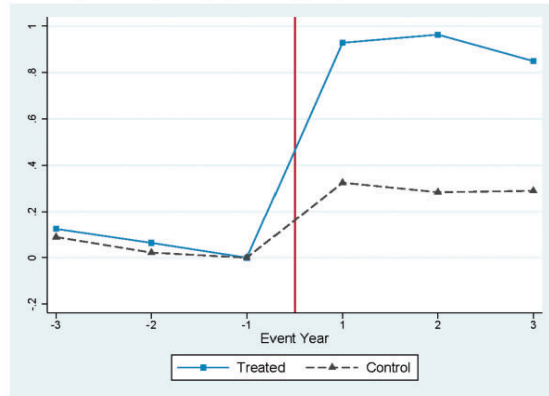
Economic message: The Moody’s effect is the first stage. The S&P placebo supports the interpretation that the recalibration is not driven by underlying credit improvements.

Table 3
Difference-in-differences of ratings around the recalibration

	Rating Moody's		Rating S&P		Rating Moody's		Rating S&P	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>A. County level</i>								
	<i>Sample of bond issuers</i>				<i>Sample of urban counties</i>			
Recalibrated $\times$ Post	0.700*** (0.162)	0.727*** (0.170)	0.150 (0.218)	0.147 (0.235)	0.590*** (0.215)	0.564*** (0.227)	-0.019 (0.241)	0.004 (0.261)
State-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Size group-year fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.31	0.33	0.25	0.30	0.29	0.31	0.25	0.29
Number of observations	4,872	4,872	2,147	2,147	3,168	3,168	1,918	1,918
Number of counties	1,319	1,319	608	608	701	701	485	485
<i>B. Local government level</i>								
	<i>Sample of bond issuers</i>				<i>Sample of urban counties</i>			
Recalibrated dummy $\times$ Post	0.653*** (0.067)	0.666*** (0.076)	0.114 (0.199)	0.167 (0.289)	0.663*** (0.071)	0.666*** (0.077)	0.215 (0.224)	0.167 (0.289)
Local gov. type-year fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
County-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Local gov. fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.53	0.61	0.31	0.40	0.52	0.60	0.33	0.40
Number of observations	11,258	11,258	4,567	4,567	8,941	8,941	4,094	4,094
Number of local governments	4,672	4,672	1,720	1,720	3,611	3,611	1,541	1,541

This table presents difference-in-differences estimates of regressions of Moody's and S&P ratings around the Moody's recalibration in April–May 2010. Panel A presents county-level results using the average rating across all issues of local governments of each county and event year. *Recalibrated* is the fraction of local government units upgraded in each county during the Moody's recalibration. Panel B presents local-government-level results using the average rating across all issues of each local government and event year. *Recalibrated dummy* takes a value of one if a local government experienced an upgrade of any of its outstanding bonds during the Moody's recalibration. *Post* is a dummy variable that takes a value of one between April 2010 and March 2013. Controls include a dummy for general obligation bonds, a dummy for Build America Bonds, and duration. The sample in Columns (1)–(4) consists of counties in which at least one local government issued bonds in the municipal bond market during the four-year period before the recalibration (April 2006–March 2010). The sample in Columns (5)–(8) consists of urban counties. The sample period is from April 2006 to March 2013. Robust standard errors clustered at the county level (in panel A) and local government level (in panel B) are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

A Change in Moody's ratings (relative to year -1)



B Difference between treatment and control groups

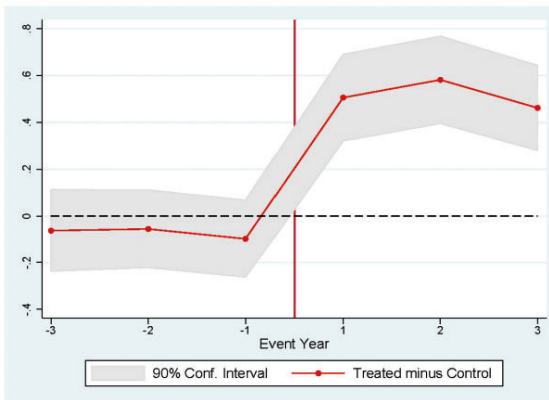


Figure 2
Moody's Ratings around the recalibration
This figure shows the regression coefficients of Moody's ratings on event-year dummies around the Moody's recalibration event from April to May 2010. Panel A presents the change in Moody's rating (relative to year -1) for treated and control groups. Panel B presents the difference between treatment and control groups.

Figure 2: Moody's ratings around recalibration

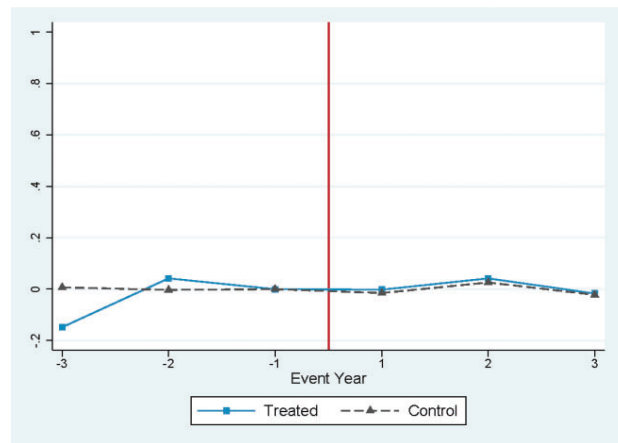


Figure 3
S&P ratings around the recalibration
This figure shows the regression coefficients of S&P's ratings on event-year dummies around the Moody's recalibration event from April to May 2010. The figure presents the change in Moody's rating (relative to year -1) for treated and control groups.

Figure 3: S&P ratings around recalibration

5.4 Table 4 and Figure 4: bond issuance and offer yields

Read:

- Table 4 shows that treated local governments issue more bonds after recalibration.
- At the local-government level, issue amount increases by about 16–20% relative to controls.
- Offer yields fall for upgraded governments. The local-government-level reduction is about 13–14 basis points.
- Figure 4 shows that treated counties' issue amounts rise relative to control counties after the event.

Economic message: Ratings matter in the primary municipal bond market. The recalibration relaxes financing constraints by raising debt capacity and reducing borrowing costs.

Table 4
Difference-in-differences of issue amount and offer yield around the recalibration

	Issue amount (log)		Offer yield		Issue amount (log)		Offer yield	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. County level								
	Sample of bond issuers				Sample of urban counties			
Recalibrated×Post	0.334*** (0.093)	0.289*** (0.097)	-0.454*** (0.140)	-0.614*** (0.146)	0.407*** (0.131)	0.408*** (0.137)	-0.591*** (0.221)	-0.600*** (0.228)
State-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Size group-year fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.22	0.24	0.29	0.31	0.32	0.33	0.39	0.41
Number of observations	6,843	6,843	6,843	6,843	3,912	3,912	3,912	3,912
Number of counties	1,859	1,859	1,859	1,859	799	799	799	799
B. Local government level								
	Sample of bond issuers				Sample of urban counties			
Recalibrated dummy×Post	0.200*** (0.046)	0.164*** (0.047)	-0.141** (0.060)	-0.127** (0.061)	0.206*** (0.048)	0.174*** (0.050)	-0.162** (0.064)	-0.139* (0.065)
Local gov. type-year fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
County-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Local gov. fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.30	0.31	0.53	0.54	0.29	0.30	0.52	0.53
Number of observations	18,215	18,215	18,215	18,215	13,982	13,982	13,982	13,982
Number of local governments	8,114	8,114	8,114	8,114	5,916	5,916	5,916	5,916

This table presents difference-in-differences estimates of regressions of the logarithm of the *Issue Amount* and *Offer Yield* around the Moody's recalibration in April–May 2010. Panel A presents county-level results using the logarithm of the amount of bonds issued and the average offer yield across all issues of local governments in each county and event year. *Recalibrated* is the fraction of local government units upgraded in each county during the Moody's recalibration. Panel B presents local-government-level results using the logarithm of the amount of bonds issued and the average offer yield across all issues of each local government and event year. *Recalibrated dummy* takes a value of one if a local government experienced an upgrade of any of its outstanding bonds during the Moody's recalibration. *Post* is a dummy variable that takes a value of one between April 2010 and March 2013. Controls include a dummy for general obligation bonds, a dummy for Build America Bonds, and duration. The sample in Columns (1)–(4) consists of counties in which at least one local government issued bonds in the municipal bond market during the four-year period before the recalibration (April 2006–March 2010). The sample in Columns (5)–(8) consists of urban counties. The sample period is from April 2006 to March 2013. Robust standard errors clustered at the county level (in panel A) and local government level (in panel B) are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

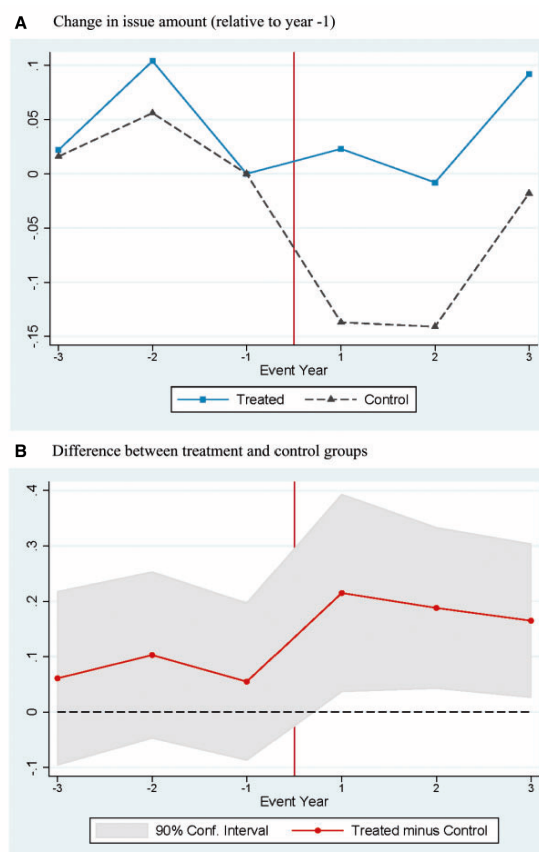


Figure 4
Issue amount around the recalibration
This figure shows the regression coefficients of logarithm of the issue amount on event-year dummies around the Moody's recalibration event from April to May 2010. Panel A presents the change in logarithm of the

5.5 Table 5 and Figure 5: local government expenditures

Read:

- Table 5 shows that local government expenditures rise after recalibration.
- In the county-level bond-issuer sample, coefficients range from 0.045 to 0.061 in the 2007–2013 panel, and are statistically significant.
- In the urban sample, coefficients are larger, around 0.071 to 0.095 in the 2007–2013 panel.

- A one-standard-deviation increase in recalibration exposure implies roughly a 0.5% increase in expenditures.
- Figure 5 shows treated and control expenditure paths are similar before the event, while treated counties rise more after.

Economic message: The upgraded ratings translate into higher public spending, which is the key fiscal channel.

Table 5
Difference-in-differences of local government expenditures around the recalibration

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Panel 2007–2013		Panel 2009–2012		Panel 2007–2013		Panel 2009–2012	
A. County level	Sample of bond issuers				Sample of urban counties			
Recalibrated × Post	0.061** (0.029)	0.045* (0.027)	0.053** (0.026)	0.027 (0.026)	0.095*** (0.027)	0.071** (0.029)	0.078*** (0.025)	0.044* (0.026)
State-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Size group-year fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.39	0.39	0.12	0.13	0.47	0.48	0.18	0.20
Number of observations	12,159	12,159	6,948	6,948	6,349	6,342	3,628	3,624
Number of counties	1,737	1,737	1,737	1,737	907	906	907	906
B. Local government level	Sample of bond issuers				Sample of urban counties			
Recalibrated dummy × Post	0.007 (0.010)	0.019 (0.015)	0.022** (0.009)	0.029** (0.013)	0.023** (0.011)	0.028* (0.016)	0.029*** (0.010)	0.038*** (0.014)
Local gov. type-year fixed effects	Yes	No	Yes	No	Yes	No	Yes	No
County-year fixed effects	Yes	No	Yes	No	Yes	No	Yes	No
Local gov. type-county-year fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
Local gov. fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.28	0.39	0.24	0.34	0.24	0.35	0.19	0.31
Number of observations	21,430	15,054	14,286	9,725	17,756	12,710	11,836	8,222
Number of local governments	3,572	2,726	3,572	2,466	2,960	2,271	2,960	2,080

This table presents difference-in-differences estimates of regressions of the logarithm of local government expenditures (as of July of each year) around the Moody's recalibration in April–May 2010. Panel A presents county-level results using aggregated expenditures across all local governments in each county and year. *Recalibrated* is the fraction of local government units upgraded in each county during the Moody's recalibration. Panel B presents local-government-level results. *Recalibrated Dummy* takes a value of one if a local government experienced an upgrade of any of its outstanding bonds during the Moody's recalibration. *Post* is a dummy variable that takes a value of one in 2011 and for each year thereafter. The sample in Columns (1)–(4) consists of counties in which at least one local government issued bonds in the municipal bond market during the four-year period before the recalibration (April 2006–March 2010). The sample in Columns (5)–(8) consists of urban counties. Controls include house price index and number of households. Robust standard errors clustered at the county level (in panel A) and local government level (in panel B) are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

government expenditures as the dependent variable. Panel A shows results at the county level, and panel B shows results at the local government level. Columns (1) and (2) present the results using the 2007–2013 period, and

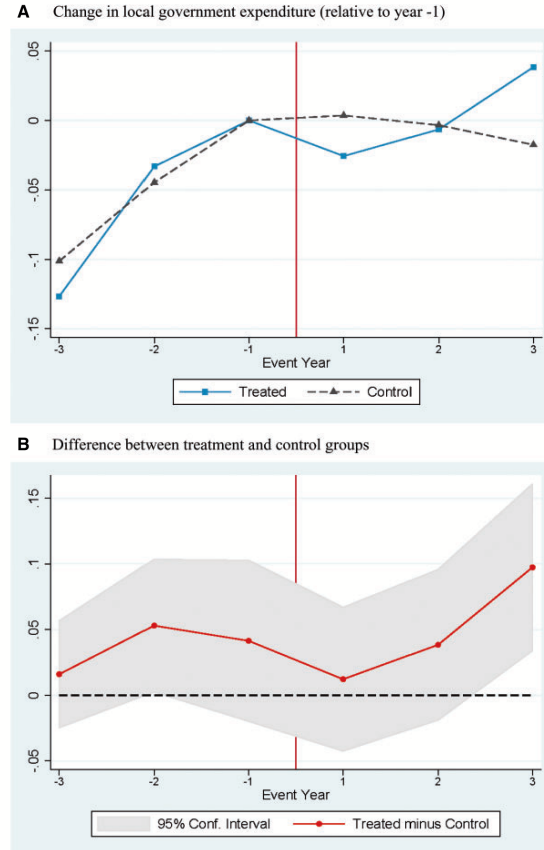


Figure 5
Local government expenditures around the recalibration
This figure shows the regression coefficients of logarithm of local government expenditures (as of July of each year) on event-year dummies around the Moody's recalibration event from April to May 2010. Panel A presents

5.6 Table 6 and Figure 6: local government employment

Read:

- County-level local government employment rises after recalibration.
- In the bond-issuer sample, the coefficient is 0.039 to 0.057 in the 2007–2013 panel.
- In the urban-county sample, the coefficient is about 0.075 in the 2007–2013 panel.
- Local-government-level employment effects are smaller and less precisely estimated.
- Figure 6 shows a post-event divergence between treated and control local government employment paths.

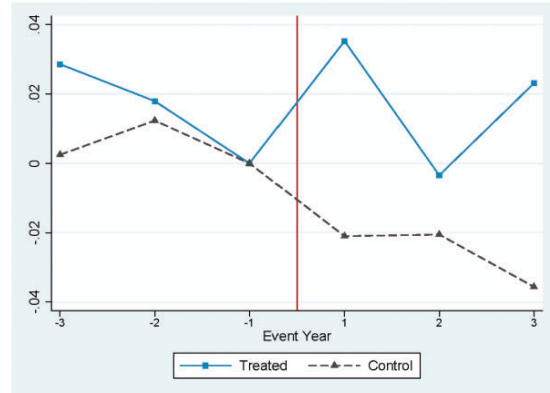
Economic message: Increased bond-financed expenditure is associated with more public-sector employment, though county-level aggregation captures the effect more clearly than individual local-government regressions.

Table 6
Difference-in-differences of local government employment around the recalibration

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Panel 2007–2013		Panel 2009–2012		Panel 2007–2013		Panel 2009–2012	
<i>A. County level</i>	<i>Sample of bond issuers</i>				<i>Sample of urban counties</i>			
Recalibrated × Post	0.039* (0.023)	0.057** (0.026)	0.052** (0.025)	0.059** (0.024)	0.075*** (0.025)	0.075*** (0.026)	0.075*** (0.023)	0.072*** (0.024)
State-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Size group-year fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.13	0.13	0.13	0.13	0.19	0.20	0.20	0.21
Number of observations	8,067	8,067	4,611	4,611	5,300	5,293	3,029	3,025
Number of counties	1,154	1,154	1,154	1,154	758	757	758	757
<i>B. Local government level</i>	<i>Sample of bond issuers</i>				<i>Sample of urban counties</i>			
Recalibrated dummy × Post	0.006 (0.012)	0.018 (0.016)	0.002 (0.010)	0.006 (0.013)	0.007 (0.012)	0.018 (0.016)	0.004 (0.010)	0.006 (0.013)
Local gov. type-year fixed effects	Yes	No	Yes	No	Yes	No	Yes	No
County-year fixed effects	Yes	No	Yes	No	Yes	No	Yes	No
Local gov. type-county-year fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
Local gov. fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.34	0.41	0.31	0.41	0.32	0.40	0.29	0.39
Number of observations	9,281	4,675	5,303	2,671	8,917	4,633	5,095	2,647
Number of local governments	1,326	668	1,326	668	1,274	662	1,274	662

This table presents difference-in-differences estimates of regressions of the logarithm of local government employment (as of March of each year) around the Moody's recalibration from April to May 2010. Panel A presents county-level results using aggregated government employment across all local governments in each county and year. *Recalibrated* is the fraction of local government units upgraded in each county during the Moody's recalibration. Panel B presents local-government-level results. *Recalibrated Dummy* takes a value of one if a local government experienced an upgrade of any of its outstanding bonds during the Moody's recalibration. *Post* is a dummy variable that takes a value of one in 2011 and for each year thereafter. The sample in Columns (1)–(4) consists of counties in which at least one local government issued bonds in the municipal bond market during the four-year period before the recalibration (April 2006–March 2010). The sample in Columns (5)–(8) consists of urban counties. Controls include house price index and number of households. Robust standard errors clustered at the county level (in panel A) and local government level (in panel B) are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

A Change in local government employment (relative to year -1)



B Difference between treatment and control groups

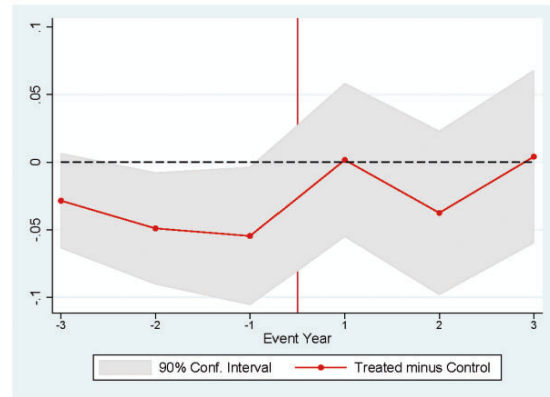


Figure 6
Local government employment around the recalibration

This figure shows the regression coefficients of logarithm of local government employment (as of March of each year) on event-year dummies around the Moody's recalibration event from April to May 2010. Panel A

5.7 Table 7 and Figures 7–8: private employment spillovers

Read:

- Table 7 shows that private employment rises after the recalibration.
- In the bond-issuer sample, the 2007–2013 coefficient is 0.059 with state-year fixed effects and 0.037 with size-year controls.
- The shorter 2009–2012 window still shows positive and statistically significant effects.
- The urban-county sample also shows positive effects, especially in the full 2007–2013 panel.
- Figure 7 shows that private employment diverges gradually after the recalibration.
- Figure 8 uses quarterly data and shows the same broad pattern at higher frequency.

Economic message: Public financing has private-sector spillovers. The delayed expansion in private employment is consistent with government spending taking time to work through local demand.

Table 7
Difference-in-differences of private employment around the recalibration

	(1)	(2)	(3)	(4)
	Panel 2007–2013		Panel 2009–2012	
<i>A. County level, sample of bond issuers</i>				
Recalibrated × Post	0.059*** (0.017)	0.037** (0.016)	0.032** (0.013)	0.021* (0.013)
State-year fixed effects	Yes	Yes	Yes	Yes
Size group-year fixed effects	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes
R-squared	0.26	0.27	0.08	0.09
Number of observations	12,331	12,331	7,050	7,050
Number of counties	1,765	1,765	1,765	1,765
<i>B. County level, sample of urban counties</i>				
Recalibrated × Post	0.041*** (0.016)	0.040*** (0.015)	0.014 (0.012)	0.021** (0.010)
State-year fixed effects	Yes	Yes	Yes	Yes
Size group-year fixed effects	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes
R-squared	0.38	0.42	0.17	0.27
Number of observations	6,508	6,501	3,718	3,714
Number of counties	930	929	930	929

This table presents difference-in-differences estimates of regressions of the logarithm of private employment (as of March of each year) in each county and year around the Moody's recalibration in April–May 2010. *Recalibrated* is the fraction of local government units upgraded in each county during the Moody's recalibration. *Post* is a dummy variable that takes a value of one in 2011 and for each year thereafter. Controls include house price index and number of households. The sample in panel A consists of counties in which at least one local government issued bonds in the municipal bond market during the four-year period before the recalibration (April 2006–March 2010). The sample in panel B consists of urban counties. Robust standard errors clustered at the county level are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level.

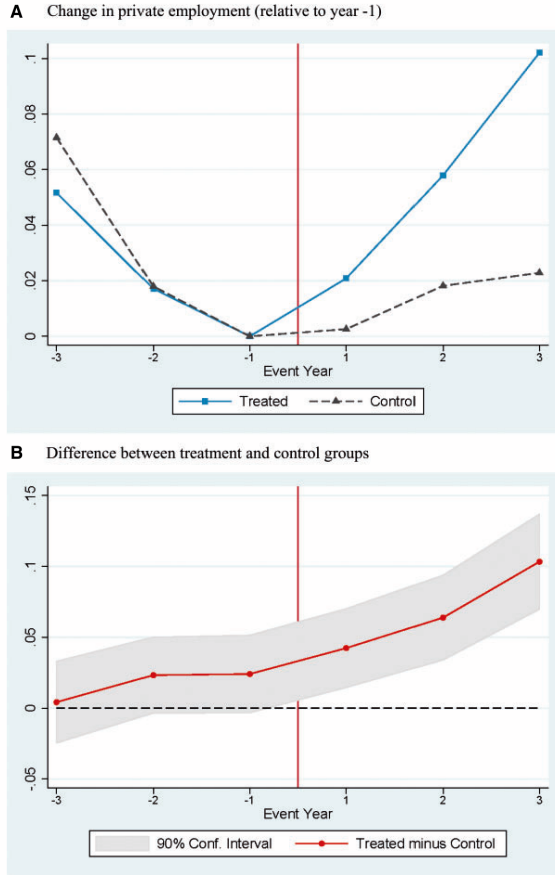


Figure 7
Private employment around the recalibration
This figure shows the regression coefficients of logarithm of private employment (as of March of each year) on event-year dummies around the Moody's recalibration event from April to May 2010. Panel A presents the change in private employment around the recalibration event from April to May 2010. Panel B presents the difference between treatment and control groups around the recalibration event from April to May 2010.

Figure 7: Annual private employment

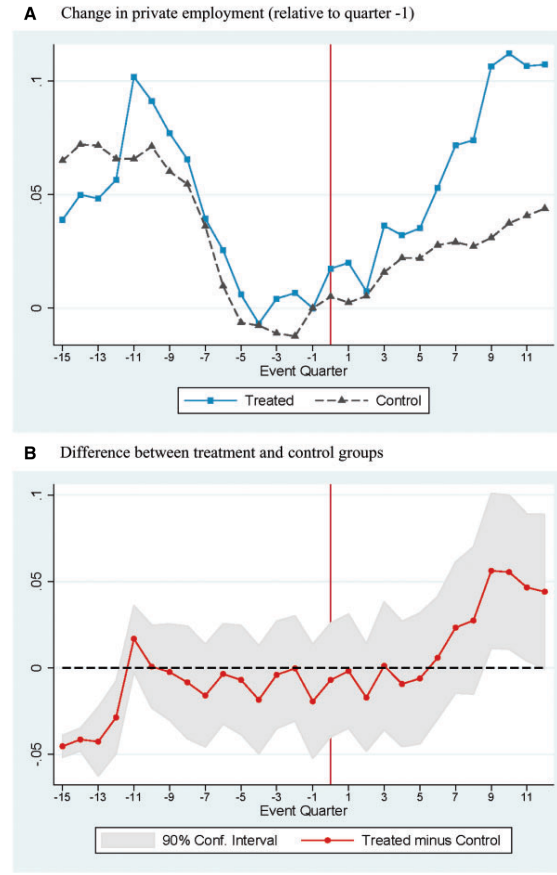


Figure 8
Private employment around the recalibration: Quarterly frequency
This figure shows the regression coefficients of logarithm of private employment (at the end of each quarter) on event-quarter dummies around the Moody's recalibration event from April to May 2010. Panel A presents the change in private employment around the recalibration event from April to May 2010. Panel B presents the difference between treatment and control groups around the recalibration event from April to May 2010.

Figure 8: Quarterly private employment

5.8 Table 8: sectoral employment

Read:

- Tradable employment effects are statistically insignificant.
- Nontradable employment rises: the bond-issuer sample shows coefficients of 0.095 and 0.065.
- Education employment rises: the bond-issuer sample shows coefficients of 0.107 and 0.068.
- Construction employment is not the dominant source of the employment response in the main table.

Economic message: The sector results support a local-demand channel. Public financing boosts sectors tied to local spending and public services rather than sectors tied to external demand.

Table 8
Difference-in-differences of private employment by sector around the recalibration

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Tradable		Nontradable		Construction		Education	
<i>A. County level, sample of bond issuers</i>								
Recalibrated $\times$ Post	−0.023 (0.144)	0.039 (0.161)	0.095*** (0.034)	0.065* (0.036)	0.001 (0.039)	0.017 (0.041)	0.107*** (0.041)	0.068* (0.040)
State-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Size group-year fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.06	0.07	0.03	0.05	0.22	0.22	0.12	0.13
Number of observations	4,670	4,670	7,025	7,025	6,469	6,469	3,480	3,480
Number of counties	1,244	1,244	1,762	1,762	1,675	1,675	1,675	1,675
<i>B. County level, sample of urban counties</i>								
Recalibrated $\times$ Post	−0.067 (0.140)	−0.085 (0.147)	0.057* (0.034)	0.033 (0.036)	0.024 (0.034)	0.020 (0.037)	0.083** (0.040)	0.054 (0.045)
State-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Size group-year fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.08	0.09	0.09	0.17	0.36	0.37	0.15	0.16
Number of observations	3,229	3,227	3,717	3,715	3,659	3,659	2,722	2,717
Number of counties	830	829	930	929	923	923	708	707

This table presents difference-in-differences estimates of regressions of the logarithm of tradable (manufacturing, NAICS codes 31–33), nontradable (retail, food and accommodation, NAICS codes 44–45 and 72), construction, and education sectors employment (as of March of each year) in each county and year around the Moody's recalibration in April–May 2010. *Recalibrated* is the fraction of local government units upgraded in each county during the Moody's recalibration. *Post* is a dummy variable that takes a value of one in 2011 and for each year thereafter. Controls include house price index and number of households. The sample in panel A consists of counties in which at least one local government issued bonds in the municipal bond market during the four-year period before the recalibration (April 2006–March 2010). The sample in panel B consists of urban counties.

5.9 Table 9 and Figure 9: income

Read:

- Table 9 shows that county income increases after recalibration.
- In the bond-issuer sample, the 2006–2012 coefficients are 0.060 and 0.028.
- The shorter 2008–2011 window gives coefficients of 0.049 and 0.029.
- Urban-county estimates are smaller but generally positive and statistically significant.
- Figure 9 shows a clear post-event gap in income between treated and control counties.

Economic message: The employment gains are accompanied by higher wages and salaries, not just a reshuffling of jobs.

Table 9

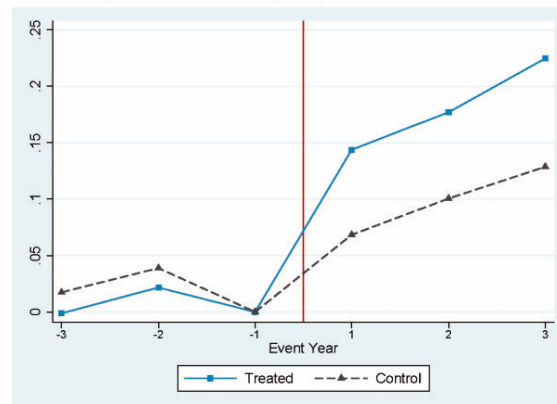
Difference-in-differences of income around the recalibration

	(1)	(2)	(3)	(4)
	Panel 2006-2012		Panel 2008-2011	
<i>A. County level, sample of bond issuers</i>				
Recalibrated $\times$ Post	0.060*** (0.017)	0.028* (0.016)	0.049*** (0.014)	0.029** (0.014)
State-year fixed effects	Yes	Yes	Yes	Yes
Size group-year fixed effects	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes
R-squared	0.67	0.67	0.72	0.72
Number of observations	10,614	10,614	7,076	7,076
Number of counties	1,769	1,769	1,769	1,769
<i>B. County level, sample of urban counties</i>				
Recalibrated $\times$ Post	0.036*** (0.009)	0.017* (0.009)	0.029** (0.014)	0.010 (0.014)
State-year fixed effects	Yes	Yes	Yes	Yes
Size group-year fixed effects	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes
R-squared	0.77	0.77	0.84	0.84
Number of observations	5,580	5,574	3,720	3,716
Number of counties	930	929	930	929

This table presents difference-in-differences estimates of regressions of the logarithm of income (as of December of each year) in each county and year around the Moody's recalibration in April–May 2010. *Recalibrated* is the fraction of local government units upgraded in each county during the Moody's recalibration. *Post* is a dummy variable that takes a value of one in 2010 and for each year thereafter. Controls include house price index and number of households. The sample in panel A consists of counties in which at least one local government issued bonds in the calendar year ending before the recalibration. Before the recalibration of April–May 2010, 1,769 counties in the sample had issued bonds in the preceding year.

Table 9: Income around recalibration

A Change in income (relative to year -1)



B Difference between treatment and control groups

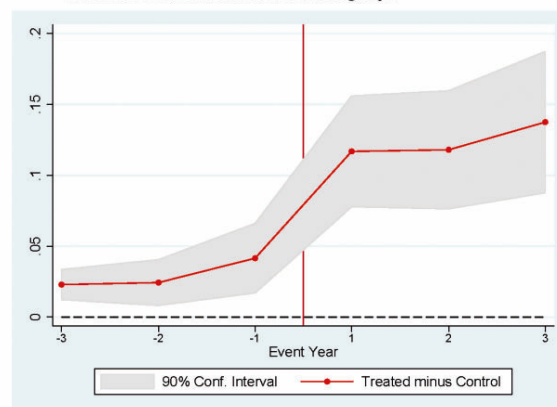


Figure 9

Income around the recalibration

This figure shows the regression coefficients of logarithm of income (as of December of each year) on event-year dummies around the Moody's recalibration event from April to May 2010. Panel A presents the change in

Figure 9: Income event-study path

5.10 Table 10: fiscal multiplier estimates

Read:

- Table 10 uses the recalibration as an instrument for local government expenditures.
- In the urban-county sample, government spending has a coefficient of 0.963 on government employment and 0.440 on private employment without size-year fixed effects.
- Income coefficients are 0.326 and 0.272 in the urban sample.
- The first-stage F -statistics in the urban sample are above ten.
- The paper converts these elasticities into about 51 jobs per \$1 million of local government spending, of which 45 are private-sector jobs.
- The implied income multiplier is about 1.9.

Economic message: The reduced-form real effects are economically meaningful. The local multiplier estimates are large but consistent with recession-era fiscal multipliers.

Table 10
Instrumental variable estimates of the elasticity of employment and income

	(1)	(2)	(3)	(4)	(5)	(6)
	Government employment		Private employment		Income	
i. County level, sample of bond issuers						
Local gov. expenditures	0.648 (0.474)	1.336* (0.779)	1.195*** (0.393)	1.026* (0.535)	0.912*** (0.324)	0.634* (0.377)
State-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Size decile-year fixed effects	No	Yes	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Chi-square statistic of instrument (first stage)	5.34	3.90	10.73	4.60	9.51	4.24
Number of observations	8,046	8,046	12,135	12,135	12,166	12,166
Number of counties	1,151	1,151	1,737	1,737	1,738	1,738
j. County level, sample of urban counties						
Local gov. expenditures	0.963*** (0.368)	1.189*** (0.434)	0.440*** (0.151)	0.662*** (0.256)	0.326** (0.138)	0.272* (0.160)
State-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Size decile-year fixed effects	No	Yes	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Chi-square statistic of instrument (first stage)	12.90	11.15	24.35	11.75	12.23	11.75
Number of observations	5,300	5,293	6,348	6,341	6,349	6,342
Number of counties	758	758	907	907	907	907

This table presents instrumental variables (two-stage least-squares) estimates of regressions of the logarithm of government employment, private employment, and income in each county and year around the Moody's recalibration in April–May 2010. Local government expenditures are instrumented with the *Recalibrated* $\times$ *Post* interaction variable. *Recalibrated* is the fraction of local government units upgraded in each county during the Moody's recalibration. *Post* is a dummy variable that takes a value of one in 2011 (2010 in the case of income) and for each year thereafter. Controls include house price index and number of households. The sample in panel A consists of counties in which at least one local government issued bonds in the municipal bond market during the four-year period before the recalibration (April 2006–March 2010). The sample in panel B consists of urban

5.11 Tables 11–12: heterogeneity by slack and financial constraints

Read:

- Table 11 shows that the recalibration effect is stronger in high-unemployment counties.
- The triple interaction with high slack is positive for government expenditures, private employment, and income.
- Table 12 shows that high-leverage local governments increase expenditures more strongly after the recalibration.
- The triple interaction with high leverage is positive and statistically significant in most specifications.

Economic message: These heterogeneity tests support the paper's interpretation: fiscal effects are stronger where local economies have slack and where governments are more financially constrained.

Table 11
Difference-in-differences of economic outcomes around the recalibration: Effect of economic slack

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Government expenditures		Government employment		Private employment		Income	
<i>A. County level, sample of bond issuers</i>								
Recalibrated $\times$ Post	-0.021 (0.027)	0.008 (0.028)	-0.006 (0.031)	0.033 (0.031)	0.019 (0.016)	0.014 (0.014)	0.022 (0.017)	0.005 (0.016)
Recalibrated $\times$ Post $\times$ High slack	0.108* (0.059)	0.115** (0.058)	0.091 (0.058)	0.087 (0.057)	0.067* (0.028)	0.051* (0.030)	0.089*** (0.029)	0.099*** (0.028)
Year fixed effects	Yes	No	Yes	No	Yes	No	Yes	No
Size group-year fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.29	0.30	0.03	0.21	0.21	0.22	0.60	0.60
Number of observations	12,180	12,180	8,067	8,067	12,345	12,345	10,626	10,626
Number of counties	1,740	1,740	1,154	1,154	1,767	1,767	1,771	1,771
<i>B. County level, sample of urban counties</i>								
Recalibrated $\times$ Post	0.017 (0.028)	0.015 (0.029)	0.035 (0.029)	0.047* (0.028)	0.003 (0.031)	0.004 (0.034)	-0.005 (0.037)	-0.012 (0.036)
Recalibrated $\times$ Post $\times$ High slack	0.067 (0.057)	0.059 (0.058)	0.118* (0.061)	0.115* (0.062)	0.039 (0.034)	0.031 (0.031)	0.103*** (0.036)	0.097*** (0.036)
Year fixed effects	Yes	No	Yes	No	Yes	No	Yes	No
Size group-year fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.35	0.37	0.07	0.08	0.29	0.33	0.70	0.71
Number of observations	6,363	6,356	5,314	5,307	6,522	6,515	5,592	5,586
Number of counties	909	908	760	759	932	931	932	931

This table presents difference-in-differences estimates of regressions of the logarithm of government expenditures, government employment, private employment, and income in each county and year around the Moody's recalibration in April-May 2010. *Recalibrated* is the fraction of local government units upgraded in each county during the Moody's recalibration. *Post* is a dummy variable that takes a value of one in 2011 (2010 in the case of income) and for each year thereafter. *High Slack* is a dummy variable that takes a value of one when the county-level unemployment rate in 2010 is above the median. Controls include house price index and number of households. The sample in panel A consists of counties in which at least one local government issued bonds in the municipal bond market during the four-year period before the recalibration (April 2006–March 2010). The sample in panel B consists of urban counties. The sample period is from 2007 to 2013/2006 to 2013 in the case of income. Robust standard errors clustered at the county level are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Table 12
Difference-in-differences of local government expenditures around the recalibration: Effect of financial constraints

	(1)	(2)	(3)	(4)
	Panel 2007–2013		Panel 2009–2012	
<i>A. Local government level, sample of bond issuers</i>				
Recalibrated dummy $\times$ Post	−0.011 (0.012)	0.005 (0.014)	0.004 (0.010)	0.013 (0.013)
Recalibrated dummy $\times$ Post $\times$ High leverage	0.054** (0.023)	0.051 (0.035)	0.050*** (0.019)	0.051* (0.030)
Local gov. type-year fixed effect	Yes	No	Yes	No
County-year fixed effect	Yes	No	Yes	No
Local gov. type-county-year fixed effects	No	Yes	No	Yes
Local gov. fixed effects	Yes	Yes	Yes	Yes
R-squared	0.33	0.43	0.27	0.37
Number of observations	18,954	12,878	12,636	8,225
Number of local governments	3,159	2,372	3,159	2,086
<i>B. Local government level, sample of urban counties</i>				
Recalibrated dummy $\times$ Post	−0.014 (0.013)	−0.001 (0.015)	0.001 (0.011)	0.009 (0.014)
Recalibrated dummy $\times$ Post $\times$ High leverage	0.058** (0.023)	0.067** (0.034)	0.056*** (0.020)	0.064*** (0.029)
Local gov. type-year fixed effect	Yes	No	Yes	No
County-year fixed effect	Yes	No	Yes	No
Local gov. type-county-year fixed effects	No	Yes	No	Yes
Local gov. fixed effects	Yes	Yes	Yes	Yes
R-squared	0.31	0.41	0.26	0.36
Number of observations	15,984	10,953	10,656	7,008
Number of local governments	2,664	2,000	2,664	1,776

This table presents difference-in-differences estimates of regressions of the logarithm of expenditures (as of July of each year) of each local government unit and year around the Moody's recalibration in April–May 2010. *Recalibrated Dummy* takes a value of one if a local government experienced an upgrade of any of its outstanding bonds during the Moody's recalibration. *Post* is a dummy variable that takes a value of one in 2011 and for each year thereafter. *High Leverage* is a dummy variable that takes a value of one when the ratio of total debt-to-revenues (as of 2007) is above the median. The sample in panel A consists of local governments with at least one bond issued in the municipal bond market during the four-year period before the recalibration (April 2006–March 2010). The sample in panel B consists local government within urban counties. Robust standard errors clustered

6 Short summary

- The paper studies the real effects of public financing using Moody's 2010 municipal bond ratings recalibration.
- The recalibration upgraded many municipal bonds by zero to three notches without reflecting improved credit quality.
- The treatment variable is the fraction of local government units in a county upgraded by the recalibration.
- The main design is difference-in-differences around April–May 2010.
- Moody's ratings increase for treated governments, while S&P ratings do not.
- Upgraded local governments issue more bonds and pay lower offer yields.
- Local government expenditures rise after ratings upgrades.
- Local government employment rises as well.
- Private employment rises, indicating spillovers outside the public sector.
- Employment effects are concentrated in nontradable and education sectors.
- County income rises after the recalibration.
- IV estimates imply about 51 jobs per \$1 million in local spending.
- About 45 of those jobs are outside the public sector.
- The implied cost per job is about \$20,000 per year.
- The implied local income multiplier is about 1.9.
- Effects are stronger in high-unemployment counties, consistent with Keynesian slack.
- Effects are stronger for high-leverage local governments, consistent with financing constraints.
- The paper estimates local multipliers, not aggregate national multipliers.
- The main economic lesson is that municipal bond financing can materially affect local real outcomes, especially in recessions.

7 Conclusion

7.1 Core conclusion

Adelino, Cunha, and Ferreira show that easing municipal financing constraints can increase local government spending and generate real economic effects. The Moody's recalibration provides a rare shock to public credit supply that is not driven by changes in local fundamentals.

7.2 Economic intuition

Ratings affect borrowing capacity and offer yields in municipal bond markets. When local governments receive exogenous upgrades, they can issue more debt at lower costs. During the Great Recession period, that additional financing supports expenditures and employment. Because local economies have slack, public spending spills over into local private employment and income rather than fully crowding out private activity.

7.3 Takeaway

The paper links municipal bond markets to real local economic outcomes. Its central message is that public-sector credit constraints matter: loosening them can create jobs and raise income, especially when local governments are constrained and local labor markets have unused capacity.